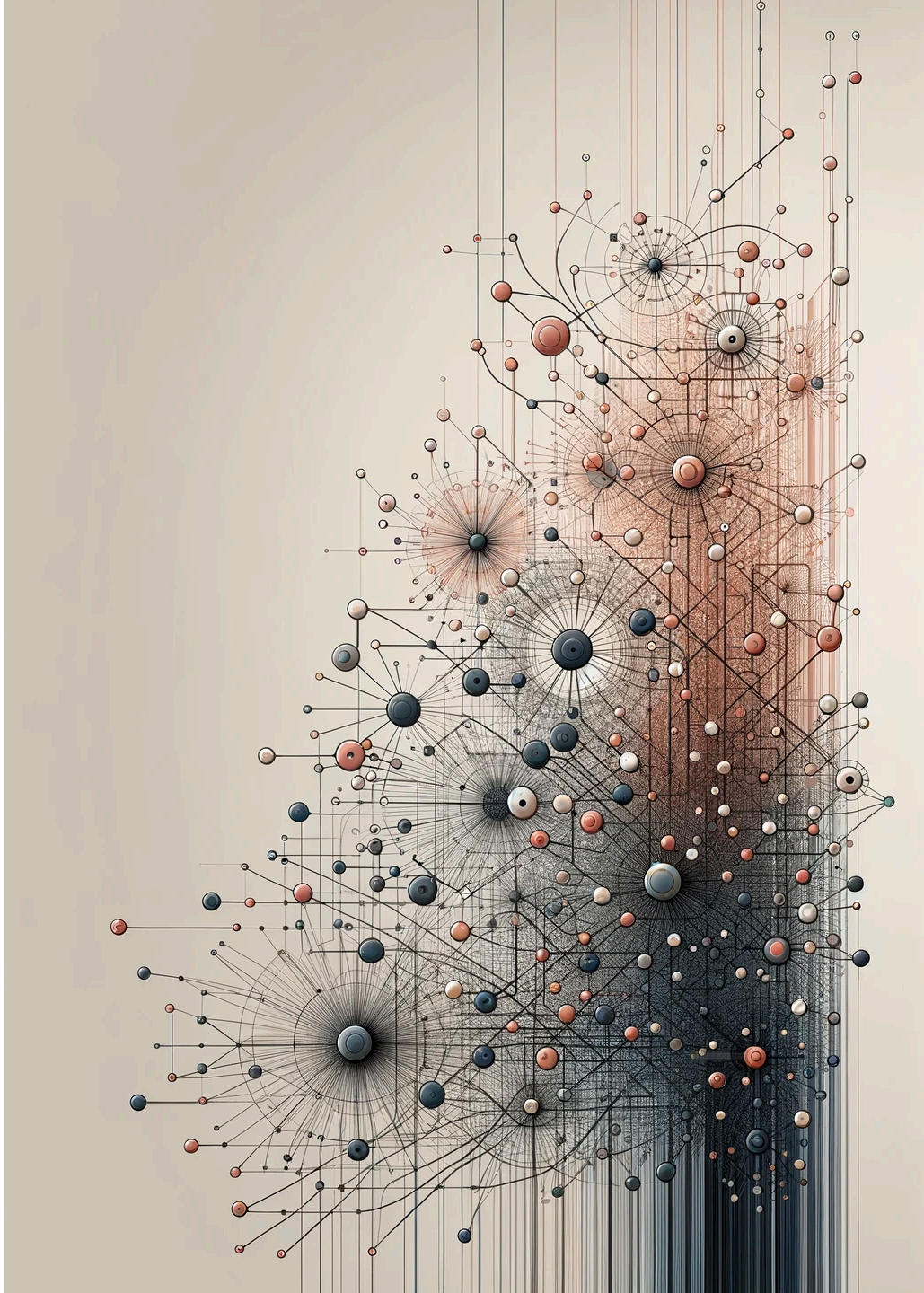
UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS17 - Traffic Networks

Selfish Routing, Braess's Paradox, and System Cost



Outline

- Traffic as a complex system
- Traffic networks as games
- Equilibrium traffic
- Social cost and social optimum
- Braess's paradox
- Best-response dynamics and potential functions
- Efficiency loss and policy interpretation

From NS16 to NS17

In `NS16`, we saw that:

- Nash equilibrium means no player wants to deviate alone
- equilibrium can be stable without being socially optimal
- incentives can create collective inefficiency

Now we put these ideas on a network.

The application is traffic.

Traffic as a Networked Strategic System

Routes, congestion, and incentives interact on the same graph.

Traffic as a complex system

Traffic is a canonical complex system.

It has:

- many agents
- local decisions
- network constraints
- congestion feedback
- emergent system-level patterns

No single driver creates congestion alone.

Congestion is an interaction effect.

Why networks matter in traffic

The road network shapes:

- which routes exist
- which alternatives are realistic
- where congestion accumulates
- whether a shortcut helps or hurts
- which equilibria are possible

Infrastructure is not just capacity.

It changes the strategic environment.

The main lesson

Traffic shows a central complex-systems theme:

local optimization can produce global inefficiency.

Each driver may choose the fastest route available to them.

But when everyone does this, the collective result can be worse than a coordinated assignment.

Braess's paradox is the sharpest example.

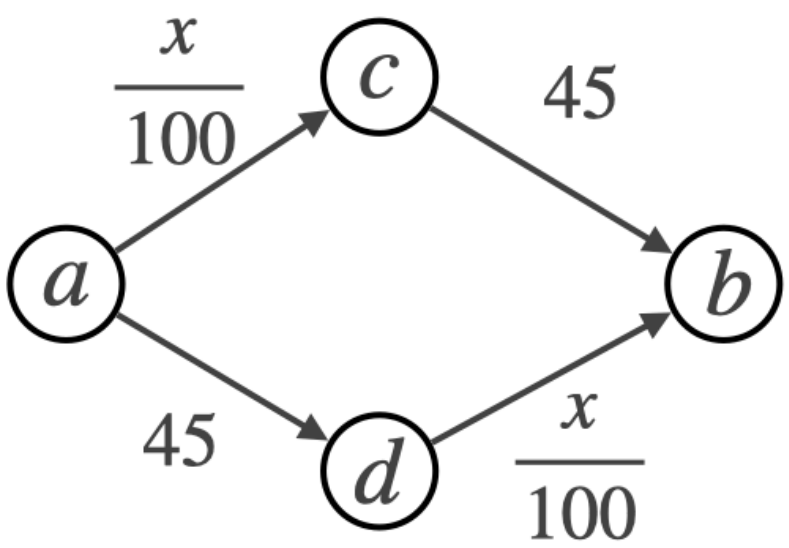
Traffic Networks as Games

Drivers choose routes, but route costs are created collectively.

Traffic network model

We model a road system as a directed graph.

- nodes are intersections, exits, or decision points
- directed edges are roads
- each edge has a travel time
- drivers travel from an origin a to a destination b



Drivers, routes, and strategies

In the traffic game:

- players are drivers
- strategies are routes from a to b
- costs are travel times
- payoffs can be interpreted as negative travel times

Drivers prefer lower cost.

So the game is usually described as cost minimization.

Edge latency functions

Each edge e has a latency function:

$$T_e(x) = a_e x + b_e$$

where:

- x is the number of drivers using edge e
- $a_e \geq 0$ is the congestion sensitivity
- $b_e \geq 0$ is the baseline travel time

Constant-time roads are allowed when $a_e = 0$.

Atomic versus continuous traffic

There are two common modeling choices.

Atomic model

- each driver is an individual player
- routes are indivisible choices

Continuous-flow model

- traffic is treated as divisible flow
- useful for very large populations

This lecture uses the atomic model for the potential-function argument.

Wardrop intuition

In large traffic systems, a useful continuous-flow intuition is Wardrop's principle:

all used routes between the same origin and destination have equal travel time.

If one used route were slower than another available route, some flow would move away from it.

This is the continuous analogue of Nash equilibrium for many small drivers.

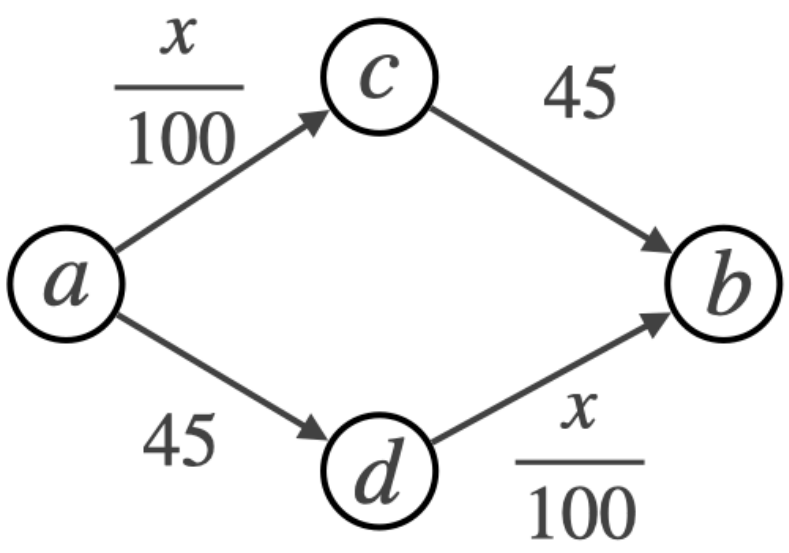
First two-route network

Suppose there are 4000 drivers traveling from a to b.

There are two routes:

- upper route: $a \to c \to b$
- lower route: $a \to d \to b$

Let x drivers use the upper route and y drivers use the lower route.



Route travel times

The two route costs are:

$$T_1(x) = \frac{x}{100} + 45$$
$$T_2(y) = 45 + \frac{y}{100}$$

The total number of drivers is:

$$x + y = 4000$$

Equilibrium traffic: intuition

At equilibrium, no driver can reduce their travel time by switching routes.

If one route is strictly faster, drivers on the slower route have an incentive to switch.

So, when both routes are used at equilibrium, their travel times must be equal.

This is the traffic version of mutual best response.

Equilibrium calculation

At equilibrium:

$$\frac{x}{100} + 45 = 45 + \frac{y}{100}$$

So:

$$x = y$$

Since $x + y = 4000$:

$$x = y = 2000$$

Every driver experiences 65 minutes.

Social cost

The **social cost** of a traffic pattern is the sum of all drivers' travel times.

If each of 4000 drivers spends 65 minutes:

$$\mathrm{SC} = 4000 \cdot 65$$

Social cost measures system-level performance.

It is not the same as one driver's travel time.

Social optimum

A traffic pattern is socially optimal if it minimizes total travel time.

Equilibrium asks:

does any individual driver want to switch?

Social optimum asks:

is total travel time as small as possible?

These two questions can have different answers.

Braess's Paradox

More network capacity can reduce performance when it changes incentives in the wrong way.

Recap: two levels of analysis

Traffic must be analyzed at two levels.

Individual level:

- each driver chooses a route
- each driver responds to travel time
- each driver minimizes personal cost

System level:

- routes share edges
- congestion creates externalities
- total travel time can be inefficient

Braess's paradox

Braess's paradox says:

adding a road can make everyone worse off.

This sounds impossible if we think only in terms of capacity.

It becomes possible when we remember that adding a road changes incentives.

The new road changes the game.

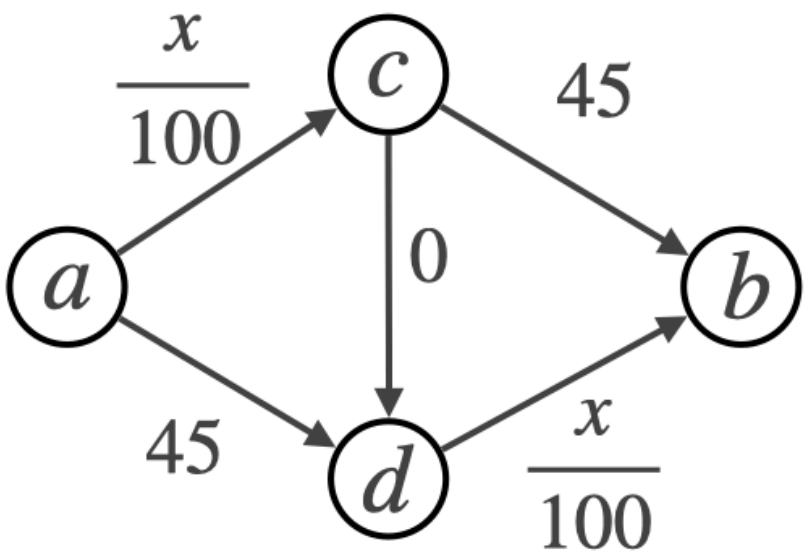
Braess network before the shortcut

Before adding the shortcut, drivers have two main routes:

- $a \to c \to b$
- $a \to d \to b$

At equilibrium, the drivers split equally.

Each route takes 65 minutes.



Before shortcut: system cost

With 4000 drivers and 65 minutes per driver:

$$\mathrm{SC}_{\mathrm{before}} = 4000 \cdot 65$$

The equilibrium is also socially good in this simple starting network.

There is no obvious conflict yet.

The paradox appears after the network changes.

Add a zero-time shortcut

Now add a new road:

$$c \to d$$

with zero travel time.

The new route is:

$$a \to c \to d \to b$$

At first glance, this should help.

More options should not hurt, right?

Why the shortcut is tempting

The shortcut connects the two congestion-sensitive edges.

A driver sees:

- $a \to c$ depends on traffic
- $c \to d$ is free
- $d \to b$ depends on traffic

Individually, the shortcut route looks attractive.

But if everyone has the same incentive, the shared edges become overloaded.

After shortcut: new equilibrium

If all 4000 drivers use the shortcut route, its travel time is:

$$\frac{4000}{100} + 0 + \frac{4000}{100} = 80$$

So every driver experiences 80 minutes.

This is worse than the original 65 minutes.

Why no driver deviates

If one driver switches away from the shortcut route:

- the alternative still uses one congested edge with about 4000 drivers
- it also uses a fixed 45-minute edge

The alternative route takes about:

$$40 + 45 = 85$$

So no individual driver wants to switch.

The bad outcome is a Nash equilibrium.

After shortcut: system cost

The new social cost is:

$$\mathrm{SC}_{\mathrm{after}} = 4000 \cdot 80$$

Every driver is worse off.

The system is worse off.

The network has more infrastructure but lower performance.

That is Braess's paradox.

What changed?

The new road did not simply add capacity.

It changed the strategic incentives.

Before:

- drivers split across two routes
- congestion was balanced

After:

- the shortcut becomes individually attractive
- everyone piles into the same strategic trap

Braess and the Prisoner's Dilemma

Braess's paradox resembles the Prisoner's Dilemma.

In both cases:

- individual incentives point toward one behavior
- collective welfare would be better under coordination
- the equilibrium is stable but inefficient

The difference is that Braess is explicitly networked.

The topology creates the strategic trap.

Real-world cases: careful interpretation

Real cities are more complex than the toy Braess network.

Still, Braess-like mechanisms appear when:

- a shortcut attracts too much traffic
- removing a road redistributes flows
- congestion feedback changes route choices
- travel demand adapts to capacity

The toy model is not a literal city.

It isolates a mechanism.

Examples in cities

Reported Braess-like cases include:

- temporary closure of 42nd Street in New York
- removal of the Cheonggyecheon expressway in Seoul
- road closure episodes in Stuttgart

The empirical lesson is cautious:

road capacity, route choice, and demand interact.

More infrastructure does not automatically imply better flow.

Policy implications

Braess's paradox motivates:

- congestion pricing
- route guidance
- demand management
- public transport alternatives
- careful simulation before infrastructure expansion

The goal is not always to add capacity.

Sometimes the goal is to reshape incentives.

From One Paradox to a General Model

The Braess example shows the phenomenon; congestion games explain the mechanism.

Congestion games

Traffic routing is a **congestion game**.

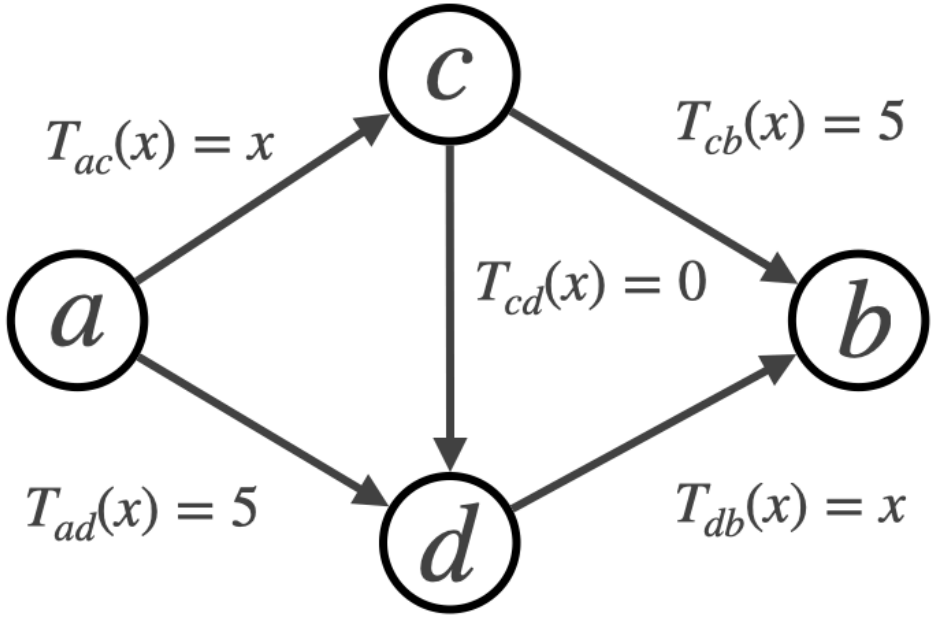
In a congestion game:

- players choose subsets of shared resources
- resource costs increase with use
- each player's cost is the sum of the costs of the resources they use

Here, resources are roads and strategies are routes.

The example for the proof

Four drivers travel from a to b; edge labels show $T_e(x)$.



Reading the travel-time functions

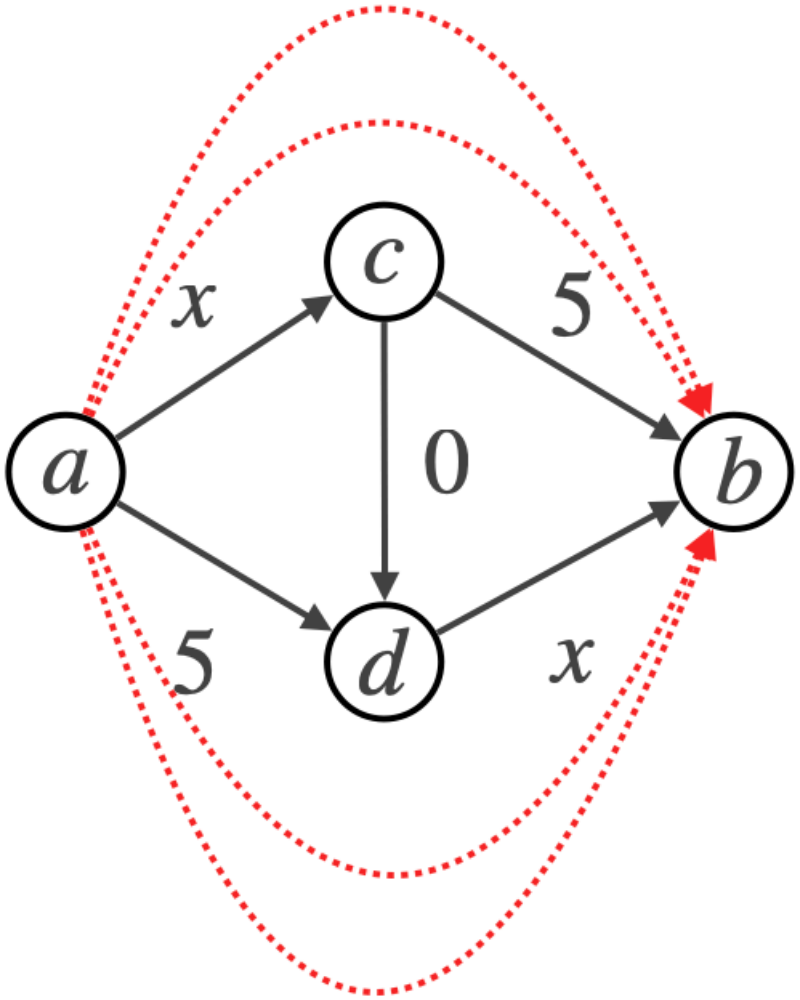
In the example:

- $T_{ac}(x) = x$: the edge gets slower as more drivers use it
- $T_{db}(x) = x$: same congestion-sensitive behavior
- $T_{ad}(x) = 5$: fixed travel time
- $T_{cb}(x) = 5$: fixed travel time
- $T_{cd}(x) = 0$: zero-time shortcut

This is a small version of the Braess network.

Traffic pattern

A **traffic pattern** assigns one complete path to each driver.



Social cost of a pattern

For a traffic pattern $\mathcal{P}$, define:

$$x_e(\mathcal{P})$$

as the number of drivers using edge e.

The total travel time on edge e is:

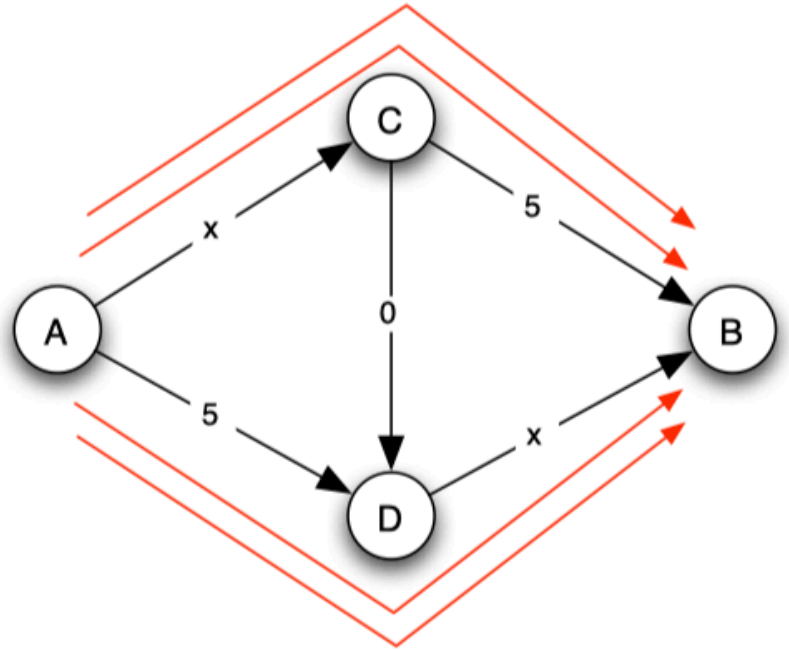
$$\mathrm{TTT}_e(\mathcal{P}) = x_e(\mathcal{P})T_e(x_e(\mathcal{P}))$$

The **social cost** is:

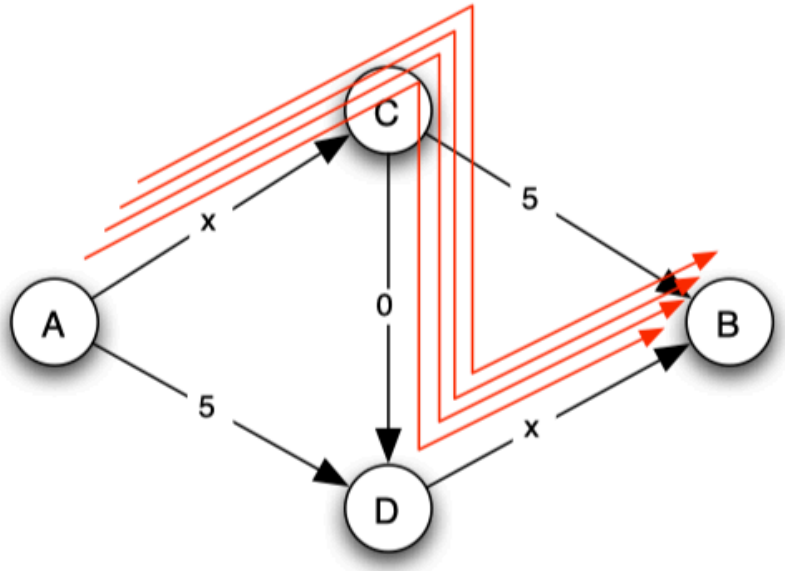
$$\mathrm{SC}(\mathcal{P}) = \sum_e x_e(\mathcal{P})T_e(x_e(\mathcal{P}))$$

Social optimum versus Nash equilibrium

Social optimum minimizes total travel time; Nash equilibrium only requires no individual profitable deviation.



(a) *The social optimum.*



(b) *The Nash equilibrium.*

Dynamics and Potential Functions

To prove that an equilibrium exists, we follow drivers as they improve their own routes.

Existence of equilibrium

The next theoretical question is:

does a traffic equilibrium always exist?

For the atomic traffic model with the latency functions used here, we can prove existence using a progress measure.

That progress measure is called a **potential function**.

Nash equilibrium in traffic

A traffic pattern $\mathcal{P}$ is a Nash equilibrium if no driver can reduce their travel time by switching to another route.

Equivalently:

each driver's current route is a best response to the routes chosen by all other drivers.

This is the many-player version of Nash equilibrium from `NS16`.

Best-response dynamics

Best-response dynamics is a simple local-update process.

1. Start from any traffic pattern.
2. If no driver can improve, stop.
3. Otherwise, choose a driver who can improve.
4. Move that driver to a strictly faster route.
5. Repeat.

If this process stops, it stops at a Nash equilibrium.

Does best-response dynamics stop?

Not every game has a pure-strategy equilibrium.

For example, Matching Pennies can cycle forever if players only use pure strategies.

So we need to prove something special about traffic games:

best-response dynamics cannot keep improving forever.

That is what the potential function will show.

Why social cost is not a progress measure

Could social cost be our progress measure?

No.

A driver's improving move can:

- reduce their own travel time
- increase travel time for others
- sometimes increase total social cost

So social cost does not necessarily decrease at every best-response step.

We need another quantity.

Potential energy of an edge

For an edge e with x drivers, define:

$$\text{Energy}(e) = T_e(1) + T_e(2) + \cdots + T_e(x)$$

If no driver uses the edge:

$$\text{Energy}(e) = 0$$

This is a cumulative quantity: imagine drivers entering the edge one by one.

Potential energy of a pattern

For a traffic pattern $\mathcal{P}$:

$$\text{Energy}(\mathcal{P}) = \sum_{e} \sum_{i=1}^{x_e(\mathcal{P})} T_e(i)$$

This is **not** social cost.

It is designed to track whether individual best-response moves make progress.

Why one switch changes potential

Suppose one driver moves from route p to route p'.

Think of the switch as two steps:

1. the driver leaves the old route p
2. the driver joins the new route p'

Leaving the old route **releases** potential energy.

Joining the new route **adds** potential energy.

The key question is the net change.

Energy released from the old route

Suppose the driver leaves route p.

For each edge e in p, if the old load was x_e, the edge energy drops from:

$$T_e(1) + \cdots + T_e(x_e)$$

to:

$$T_e(1) + \cdots + T_e(x_e - 1)$$

So the released energy on edge e is $T_e(x_e)$.

Energy added on the new route

Now the driver joins route p'.

For each edge e' in p', let $y_{e'}$ be the load **after the driver has left the old route, but before joining the new route**.

The edge energy rises from:

$$T_{e'}(1) + \cdots + T_{e'}(y_{e'})$$

to:

$$T_{e'}(1) + \cdots + T_{e'}(y_{e'} + 1)$$

So the added energy is $T_{e'}(y_{e'} + 1)$.

Route travel times in the switch

The released energy equals the travel time the driver had on the old route:

$$\sum_{e \in p} T_e(x_e)$$

The added energy equals the travel time the driver would have on the new route:

$$\sum_{e' \in p'} T_{e'}(y_{e'} + 1)$$

So the potential tracks exactly the driver's incentive to switch.

Net change in potential

The net change equals:

$$\Delta \text{Energy} = \text{new route travel time} - \text{old route travel time}$$

This is exactly why the potential function is useful.

It converts an individual improvement into a decrease in a global progress measure.

Why best-response dynamics stops

A driver switches only if:

$$\text{new travel time} < \text{old travel time}$$

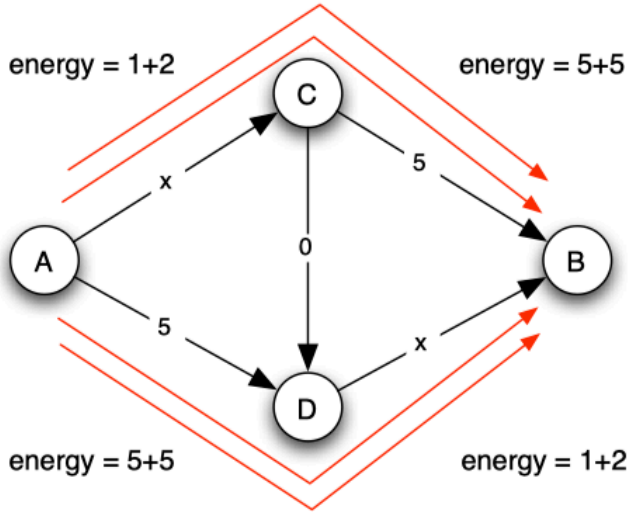
Therefore:

$$\Delta\text{Energy} < 0$$

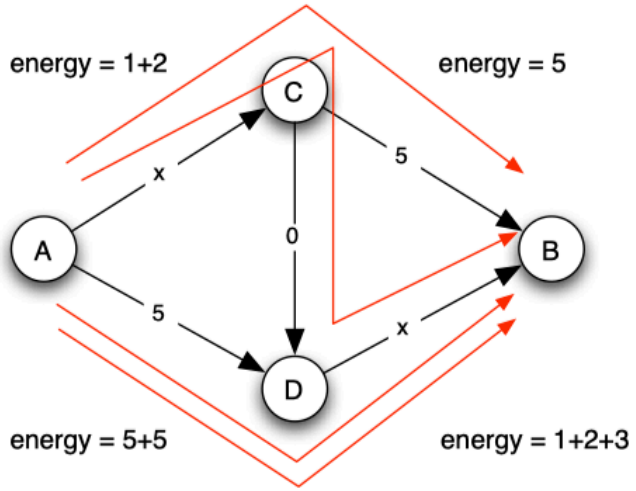
Every improving move strictly decreases potential.

There are finitely many traffic patterns.

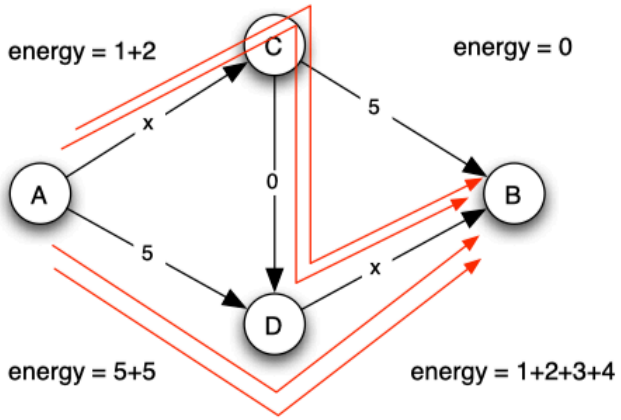
Therefore, best-response dynamics must eventually stop.



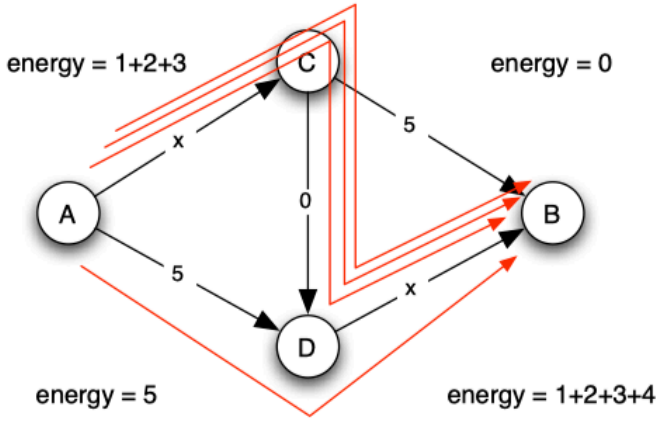
(a) *The initial traffic pattern. (Potential energy is 26.)*



(b) *After one step of best-response dynamics. (Potential energy is 24.)*



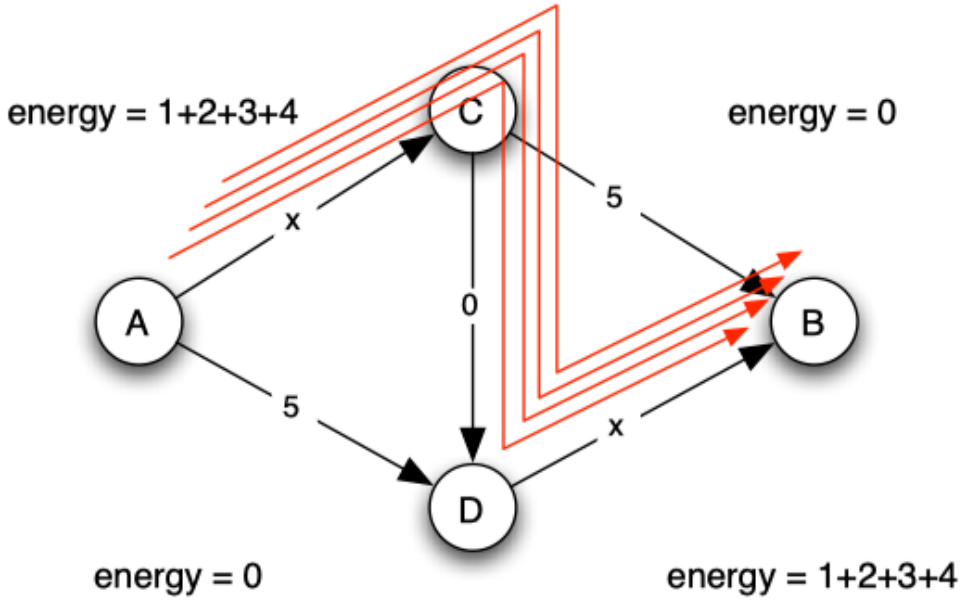
(c) *After two steps. (Potential energy is 23.)*



(d) *After three steps. (Potential energy is 21.)*

Equilibrium is reached

After four improvement steps, no driver can improve by changing route alone.



What we have proved

The potential argument proves:

- a pure Nash equilibrium exists
- best-response dynamics reaches one after finitely many improving moves
- equilibrium can emerge from decentralized updates

This is a complex-systems result:

global order can arise from local adaptation.

Potential game viewpoint

A game with this kind of progress measure is called a **potential game**.

The potential does not equal social cost.

Instead, it tracks incentives:

if one player makes a strictly improving move, the potential strictly decreases.

Traffic games have a clean equilibrium-existence argument because of this structure.

Equilibrium Quality

Existence is only the first question; the next one is how costly equilibrium can be.

Equilibrium versus optimum

Existence is not enough.

We also care about quality.

Two traffic patterns may both be equilibria, but have different social costs.

The key comparison is:

$$\text{equilibrium cost} \quad \text{versus} \quad \text{social optimum cost}$$

Potential versus total travel time

To compare equilibrium and optimum, we relate:

- potential energy
- total travel time
- social cost

For one edge with x drivers:

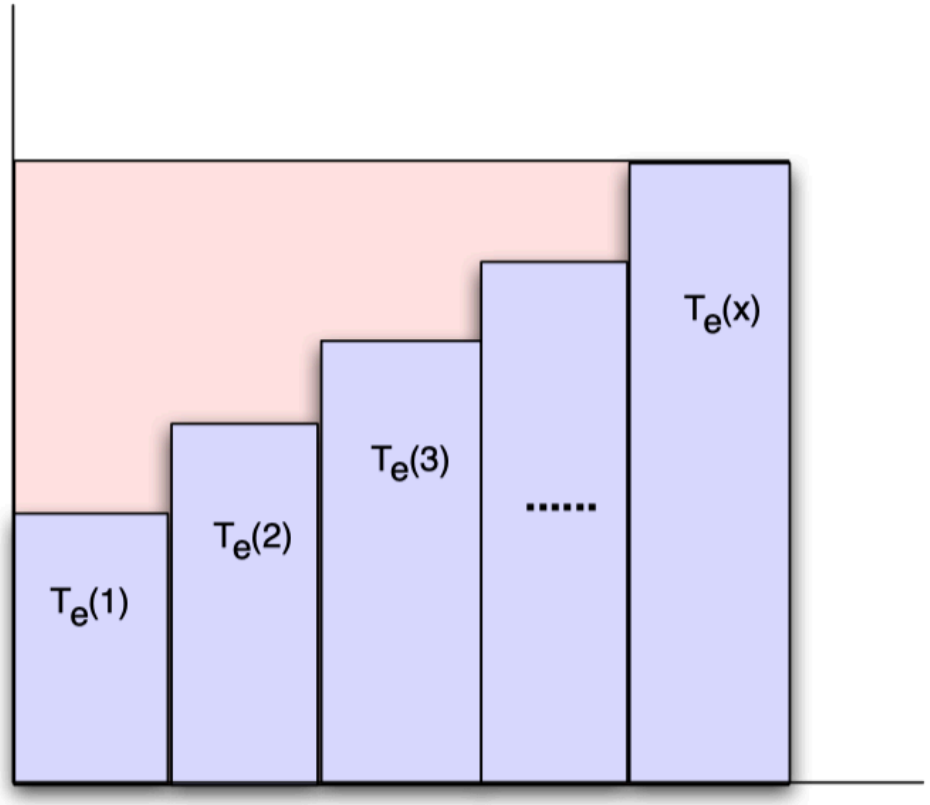
$$\mathrm{TTT}(e) = xT_e(x)$$

This is different from:

$$\mathrm{Energy}(e) = T_e(1) + \cdots + T_e(x)$$

Visual comparison

Potential energy is the blue area; total travel time is the full rectangle.



Lower bound: energy is at least half

For $T_e(x) = a_e x + b_e$:

$$\begin{aligned}
\text{Energy}(e) &= T_e(1) + T_e(2) + \cdots + T_e(x) \\
&= a_e(1 + 2 + \cdots + x) + b_e x \\
&= \frac{a_e x(x+1)}{2} + b_e x \\
&\geq \frac{1}{2} x(a_e x + b_e) \\
&= \frac{1}{2} x T_e(x) \\
&= \frac{1}{2} \text{TTT}(e)
\end{aligned}$$

Upper bound: energy is at most total travel time

Because travel time does not decrease with more drivers:

$$T_e(i) \leq T_e(x) \quad \text{for every } i \leq x$$

Therefore:

$$\text{Energy}(e) = \sum_{i=1}^{x} T_e(i) \leq \sum_{i=1}^{x} T_e(x) = xT_e(x) = \text{TTT}(e)$$

Edge-level comparison

For every edge:

$$\frac{1}{2}\mathrm{TTT}(e) \leq \mathrm{Energy}(e) \leq \mathrm{TTT}(e)$$

Summing over all edges gives the same relationship for a whole traffic pattern:

$$\frac{1}{2}\mathrm{SC}(\mathcal{P}) \leq \mathrm{Energy}(\mathcal{P}) \leq \mathrm{SC}(\mathcal{P})$$

Using the comparison

Let $\mathcal{P}_{opt}$ be a socially optimal traffic pattern.

Start best-response dynamics from $\mathcal{P}_{opt}$.

Let $\mathcal{P}_{eq}$ be the equilibrium reached by the process.

Because potential decreases along best-response dynamics:

$$\text{Energy}(\mathcal{P}_{eq}) \leq \text{Energy}(\mathcal{P}_{opt})$$

Step-by-step bound

Now chain the inequalities:

$$\begin{aligned}\mathrm{SC}(\mathcal{P}_{eq}) &\leq 2\mathrm{Energy}(\mathcal{P}_{eq}) \\ &\leq 2\mathrm{Energy}(\mathcal{P}_{opt}) \\ &\leq 2\mathrm{SC}(\mathcal{P}_{opt})\end{aligned}$$

So there exists an equilibrium whose social cost is at most twice the optimum.

This is a carefully stated existence guarantee.

What this bound does and does not say

It says:

- at least one reachable equilibrium is within a factor 2 of optimum
- potential energy helps compare local dynamics and global cost
- selfish routing has bounded inefficiency in this model

It does **not** say:

- every equilibrium is within factor 2
- this is the sharpest possible bound in every routing model
- the potential function is the same thing as social cost

Price of anarchy versus price of stability

Two related ideas are often confused.

Price of anarchy

- compares the worst equilibrium to the optimum

Price of stability

- compares the best equilibrium to the optimum

The potential proof above is closer to a best-equilibrium or existence guarantee.

Precise bounds depend on the traffic model.

Continuous routing note

In continuous-flow routing with affine latency functions, a classical bound is:

$$\mathrm{PoA} \leq \frac{4}{3}$$

This is a different model from the atomic proof used here.

The important lesson for this course is the mechanism:

selfish routing can be inefficient, and the size of inefficiency depends on modeling assumptions.

Intervention logic

The lesson is not simply "remove roads" or "add tolls".

The right question is:

how will agents respond after the intervention changes the game?

Good traffic policy evaluates the new equilibrium, not only the new infrastructure.

What to remember

Traffic networks combine:

- network topology
- strategic route choice
- congestion externalities
- equilibrium
- social cost

The surprising behavior is not a bug in the example.

It is a property of networked strategic systems.

Policy toolkit

To improve traffic, planners can change:

- the network
- the incentives
- the information available to drivers
- the demand for car trips
- the alternatives to driving

The complex-systems lesson:

interventions must account for feedback.

Main takeaway

Traffic is a network game.

Drivers optimize locally.

Congestion couples their choices.

Equilibrium can emerge without coordination.

But equilibrium can be inefficient, and even added capacity can make the system worse.

Reading material

Easley and Kleinberg, *Networks, Crowds, and Markets*

- Chapter 8: Modeling Network Traffic using Game Theory
- Sections 8.1-8.3

Recommended focus:

- traffic equilibrium
- Braess's paradox
- social cost
- potential-function argument

Related course framing: MIT OCW, *Networks*: network traffic and congestion

Q&A